ARTIFICIAL INTELLIGENCE: A RAND PERSPECTIVE

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January 1986

P-7172

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Rand Paper P-7172
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PREFACE

This paper presents a brief history of artificial intelligence (AI) research at The Rand Corporation. Rand has long been a leader in the field of AI beginning with the seminal work of Newell, Shaw, and Simon some thirty years ago and continues with recent work in expert systems and knowledge-based simulation. This paper traces the major accomplishments in AI at Rand with particular emphasis on Rand's research during the past decade. An extensive bibliography highlights the major Rand documents on AI and related subjects.

This paper was written as an introduction to the book

> *Expert Systems: Techniques, Tools, and Applications*
> by Philip Klahr and Donald A. Waterman
> Addison-Wesley, Reading, Massachusetts, 1986.

The book consists of ten Rand research reports (see Appendix) plus the historical chapter presented here.

I. THE EARLY YEARS: THE BEGINNINGS OF ARTIFICIAL INTELLIGENCE

The Rand Corporation played a major role in the early development of artificial intelligence (AI). Of the twenty chapters in the first published book on AI, the 1963 *Computers and Thought* anthology of Feigenbaum and Feldman, six had been previously published as Rand research reports [4,25,80,81,88,114]. Much of this early work in AI was the result of the collaboration of two Rand employees, Allen Newell and Cliff Shaw, and a Rand consultant, Herbert Simon of the Carnegie Institute of Technology (later to become Carnegie-Mellon University).

Beginning in the mid-1950s, Newell, Shaw, and Simon's research on the Logic Theory Machine, their chess playing program, and the General Problem Solver (GPS) defined much of the AI-related research during the first decade of AI. Their work encompassed research areas that are still prominent subfields of artificial intelligence: symbolic processing, heuristic search, problem solving, planning, learning, theorem proving, knowledge representation, and cognitive modeling. At Rand they left a legacy of publications that gave AI many of its building blocks and much of its momentum [71-91,102-104].

It is important to note that this surge of AI activity at Rand did not take place in isolation. It occurred at a time and place where a host of fundamental notions about computer science and technology were being generated. In the 1950s Rand was involved in designing and building one of the first stored-program digital computers, the JOHNNIAC[1] [35]; George Dantzig and his associates were inventing linear programming [15]; Les Ford and Ray Fulkerson were developing techniques for network flow analysis [31]; Richard Bellman was developing his ideas on dynamic programming [6]; Herman Kahn was advancing techniques for Monte Carlo simulation [48]; Lloyd Shapley was revolutionizing game

[1]The JOHNNIAC, named after John von Neumann, a Rand consultant in the late 1940s and early 1950s, was in operation from 1953 to 1966. It was used extensively by Newell, Shaw, and Simon in their work on information processing.

theory [100]; Stephen Kleene was advancing our understanding of finite automata [58]; Alfred Tarski was helping to define a theory of computation [111]; and James Culbertson [13,14] and Alton Householder [45,46] were investigating the relationship between neural nets, learning, and automata.[2]

Within this milieu, Newell, Shaw, and Simon were developing methods and directions for AI research. Perhaps equally important was their development of appropriate computational tools for AI programming. Using the notion of linked-list structures to represent symbolic information, Newell and his associates developed the first symbol-manipulating and list-processing languages, a series of IPL (Information Processing Language) languages that culminated in IPL-V [77,92]. In their 1963 paper [9], Dan Bobrow and Bert Raphael (both of MIT at the time, but also Rand consultants) included IPL-V as one of the earliest and most highly developed list-processing languages.

Because of Rand's unique computing environment and its close ties to the Carnegie Institute of Technology, several Carnegie graduate students were attracted to Rand and several Ph.D. dissertations emerged, including those of Fred Tonge [114,115] and Ed Feigenbaum [24,25]. During the early 1960s, Feigenbaum, in collaboration with Simon, continued to publish Rand reports [26-30,105] describing his experiments with his verbal learning program EPAM. Even after completing his work at Carnegie, Feigenbaum remained a Rand consultant and was highly influential in Rand's research on expert systems and expert systems languages that emerged in the early 1970s. Newell and Simon also were Rand consultants during the 1960s and 1970s. One of their associates, Don Waterman, joined Rand in the mid-1970s and brought much of their influence on the use of production systems to Rand's first work on expert systems.

But AI also had its share of controversy, at Rand as elsewhere. Given its quick rise to popularity and its ambitious claims [106], AI soon had its critics, and one of the most prominent, Hubert Dreyfus,

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[2]Much of the credit for creating this intellectually stimulating environment belongs to John Davis Williams, who led Rand's Mathematics Department in the 1950s, and also to the United States Air Force for its generous sponsorship of a broad range of research activities.

published his famous critique of AI [20] while he was consulting at Rand. In addition, the early promise of automatic machine translation of text from one language to another (the emphasis at Rand was on translation from Russian to English) produced only modest systems, and the goal of fully automated machine translation was abandoned in the early 1960s.

The research in machine translation did, however, serve to elucidate the difficult problems of automated language understanding and translation. As a result, work in this area turned more toward fundamental and generic issues of linguistic theory, and Rand engaged in over a decade of activity in computational linguistics. By 1967, Rand researchers had produced a wealth of literature (over 140 articles) on linguistic theory and research methods, computational techniques, the English and Russian languages, automatic content analysis, information retrieval, and psycholinguistics [44]. In addition, David Hays produced one of the earliest textbooks on computational linguistics [43].

During the 1960s, Rand provided a center in which natural-language researchers from all over the world could meet, communicate, and collaborate. Special seminar programs and summer symposia (e.g., [59]) provided ample opportunities for researchers to exchange ideas and test theories. Work at Rand during this period included a number of developments: Martin Kay and his associates were working on the MIND system which focused on research in morphology [52], semantic networks [53,99], and parsing [49-51]; Jane Robinson was developing new syntactic analyzers [94]; Roger Levien and Bill Maron were developing the Relational Data File for information retrieval and question answering [63-65]; Larry Kuhns was developing a sophisticated query language for database inference [60,61]; and, in a somewhat different area, work was beginning on a new theory of "fuzzy sets" [7].

II. HUMAN-ORIENTED ENVIRONMENTS

Since its early involvement with the JOHNNIAC in 1953, Rand has continually worked on the development of human-oriented interfaces. Although much of this work has been outside the AI framework, the research has provided interactive computing environments that have made AI systems easier to design, implement, debug, and understand. Today, computational environments appropriate for AI systems comprise a prominent subfield of AI research.

Major milestones in Rand's work on human-oriented environments are JOSS, the Rand Tablet, Videographics, GRAIL, and BIOMOD. Cliff Shaw's JOSS [101], one of the world's first true interactive computing systems, executed interpretively, had execution tracing facilities, could be used to solve mathematical problems with considerable precision, and had a clean, easy-to-learn syntax. These characteristics influenced a number of later programming systems and environments. JOSS has remained a key computing resource at Rand for over 20 years and is still used today.

The Rand Tablet [16] was the first example of a two-dimensional writing surface for computer communication. A capacitative coupling between a stylus and a grid of wires embedded in the tablet provided sufficient accuracy to allow recognition of hand-printed characters approximately 1/4-inch high. Entire interactive computing systems, including GRAIL and BIOMOD, were based on use of the Rand Tablet--in fact, the entire GRAIL system was programmed with the Tablet as the sole input device.

The Rand Video Graphic System [116] was one of the first interactive graphics systems to use raster-scan technology. High-resolution video displays were driven from a rotating magnetic disk. At a time when the cost of providing individual self-contained systems was prohibitive, this rapid-response system gave many users simultaneous access to precision computer graphics.

The GRAIL system [21] and its successor BIOMOD [34] pioneered the use of graphic displays and tablets as input-output devices for programming and modeling. GRAIL allowed the direct execution of

programs defined by a combination of flowcharts and coding forms, in which each box on a top-level flowchart could itself be defined in terms of another flowchart or (at the lowest level) a coding form. BIOMOD applied these interactive concepts to a major biomedical modeling system, in which transfer equations and other forms of equations describing a dynamic system could be input directly on a displayed form. The results of a simulation defined by these equations could then be viewed graphically in terms of time-history plots of any of the variables (or combinations of them). More than 15 years ago these systems used a sophisticated real-time recognition algorithm for characters hand-printed on a tablet [33]. In addition to these prototype/demonstration systems, a number of studies were conducted, principally by Barry Boehm, on the effects on user productivity of improvements in the speed and dependability of response time of interactive computer systems [96].

In 1972, several people involved in Rand's human-machine interface work, including Keith Uncapher, Tom Ellis, Bob Anderson, and Bob Balzer (who was just beginning to develop some of his ideas on program specification [5]), left Rand to form the University of Southern California's Information Sciences Institute. Anderson returned to Rand the following year and started Rand's work in an area that was to become known as "expert systems."

Rand work on interactive systems in the late 1970s centered on developing UNIX-based tools, such as the Rand Editor [8] and the MH electronic message handler [10], and on research in interactive maps for dynamically displaying geographic information [3]. Map displays continued to play a prominent role in Rand research in the 1980s, particularly the color graphic displays used to depict geographic simulations dynamically.

III. EXPERT SYSTEMS

In the early 1970s, Bob Anderson and his associates began directing their attention toward providing aids for inexperienced computer users. The objective was to enable these users to exploit the power of computers, and even to program them, without having to become computer sophisticates. At the same time, Rand researchers were becoming increasingly interested in intelligent terminals and the possibility that such terminals might eventually be developed into powerful individualized computer workstations.

One of the initial goals of these researchers was to develop a simple, English-like language for computer users who were not programmers. Such a language, combined with intelligent terminals, could bring computers to a wide range of potential users by providing an easy-to-use interactive environment in which to work.

The Rand effort was influenced by the work of Ed Feigenbaum (who retained a continuing link to Rand as a consultant) and his associates at Stanford in the early 1970s, particularly in their use of rule-based models in the development of a system that became known as MYCIN. Anderson and his associates were particularly impressed with MYCIN's explanation facilities and its very readable English-like output. MYCIN's input, however, lacked this English-like quality because it had to be programmed in LISP, a high-level language that was much too sophisticated for novice computer users. Therefore, Rand set out to build a language that allowed simple, English-like input as well as output.

That effort resulted in the RITA language [1,2].[3] RITA was designed by Rand researchers as a language for developing intelligent interfaces to computer systems. RITA's unique English-like syntax could be read fairly easily by nonprogrammers, and its control mechanism gave RITA programs easy access to the local operating system. The language was used for developing not only interface programs, but also expert

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[3]Most of Rand's work on expert systems, including RITA, ROSIE, and TATR, was supported by the Defense Advanced Research Projects Agency (DARPA) within its Information Processing Techniques Office.

systems, an application area to which RITA was not especially well suited. The problems that arose from attempts to develop expert systems in RITA (e.g., slow execution speed and the limited expressiveness of the syntax) led eventually to the development of the ROSIE language.

RITA was used with some success as a tool for heuristic modeling in studies of international terrorism [121]. This work, described in Chapter 4,[4] combined the talents of computer scientists and social scientists in the design of a knowledge-based expert system to aid terrorism researchers in their analysis of terrorist activities. The project precisely defined the basic concepts needed for a model of terrorism and used these concepts in implementing the expert system.

The success of RITA at Rand, combined with a growing interest in rule-based systems in the AI community, led two Rand researchers, Don Waterman and Rick Hayes-Roth, to organize a workshop in 1977 on rule-based pattern-directed systems. The papers presented at the workshop were later published in a book that essentially defined this emerging technology [119].

When Phil Klahr and Stan Rosenschein joined Rand's AI staff in 1978, rule-based systems became a major focus of Rand's AI research. Six Rand researchers specializing in rule-based systems gathered for an intensive two-day workshop to design the next-generation rule-based language. Although RITA had proven to be quite useful, the workshop participants wanted to develop a more sophisticated, more general-purpose, and even more English-like language specifically designed for building expert systems. The result was the first design of ROSIE (Rule-Oriented System for Implementing Expertise) [117]. Rand has since produced several versions of ROSIE, each intended to extend and improve the language [22,39,107]. The most recent version is described in Chapter 2.

ROSIE has been used in the development of several expert systems in a variety of application domains. In one application, described in Chapter 5, Rand researchers developed the Legal Decisionmaking System (LDS), a prototype expert system to assist attorneys and claims adjusters in settling product liability cases [122].[5] This system

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[4]References to the chapters in the Klahr and Waterman (1986) book are in the Appendix.

[5]LDS was funded by Rand's Institute for Civil Justice.

enabled researchers to explore the feasibility of applying knowledge-engineering techniques to the legal area. The work on legal reasoning, which initially focused on product liability in general, was later narrowed to the analysis and settlement of asbestos cases.

A second noteworthy application of ROSIE was in the area of military planning. As described in Chapter 6, a prototype expert system called TATR was developed to help targeteers select and prioritize airfields and target elements on those airfields [11]. The resulting program contained approximately 400 ROSIE rules.

Another military application of ROSIE was also underway at Rand during the early 1980s. RSAC, the Rand Strategy Assessment Center [18], was designed to provide military strategists with a war-gaming facility.[6] It combines a set of automated programs, or agents, with human teams to model superpower decisionmaking in conflict situations. RSAC used ROSIE to develop and implement the rule-based Scenario Agent, a policy-level model of nonsuperpower behavior [19,95]. The Scenario Agent reacts to a current hypothetical military situation by determining if there is a threat to a nonsuperpower and, if so, specifying actions to take (sending messages, granting overflight rights, changing alignments, etc.). ROSIE also influenced RSAC's development of RAND-ABEL, a C preprocessor that facilitates the encoding of rules and decision tables in a C-based environment [97,98]. RSAC also incorporates other AI techniques, such as scripts and goal-directed search, in its operational framework [109].

As expert systems research grew at Rand and in the AI community, Rick Hayes-Roth, Don Waterman, and Doug Lenat (of Stanford University at the time, but also a Rand consultant) organized a workshop in 1980 on rule-based systems and their application to the development of knowledge-based expert systems. This workshop produced the first comprehensive book on building expert systems [40], which includes a detailed comparison of expert-system-building languages [120]. Expert systems quickly became a prominent subfield of AI research and has provided a new set of tools for application in government and industry. Research in expert systems continues to be a primary focus of Rand's AI research activity.

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[6]RSAC has been supported by the Director of Net Assessment, Office of the Secretary of Defense.

IV. KNOWLEDGE-BASED SIMULATION

Simulation has become a powerful mechanism for helping humans understand complex phenomena. Results produced by simulations have had substantial impact on a broad range of decisions in the military, government, and industry. Unfortunately, most simulations lack the utility needed for practical applications. Simulations are costly to build, poorly organized, inadequately understood by users, difficult to modify, and poor in performance. Since the early 1960s, Rand has explored and developed techniques to make simulations more useful, understandable, modifiable, credible, and efficient.

Much of Rand's research in simulation methodology in the 1960s revolved around the development of the SIMSCRIPT language [66] and its successor, SIMSCRIPT II [54]. More recently a research group headed by Phil Klahr has focused on applying artificial intelligence and expert-systems technology to simulation.[7] The goal has been to develop a research environment that helps users build and refine simulations with which to analyze and evaluate various outcomes. The primary result of this work has been ROSS, an English-like, object-oriented simulation language [67,68]. ROSS, described in Chapter 3, provides a programming environment in which users can conveniently design, test, and modify large knowledge-based simulations of complex mechanisms.

Simulations written in ROSS are expert systems: they embody a human expert's knowledge of the objects that comprise the simulation domain. To build a ROSS simulation, it is necessary to specify the domain objects, their attributes, and their behavioral rules. ROSS has been used to design and build several military simulation systems, including a strategic air battle simulation called SWIRL (discussed briefly in Chapter 3, more thoroughly in [57]) and a tactical ground-based combat simulation called TWIRL [55]. The TWIRL system, described in Chapter 7, simulates a ground combat engagement between two opposing military forces. It includes troop deployment, artillery firing, air

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[7]Rand's research on knowledge-based simulation, including work on ROSS, SWIRL, and TWIRL has been supported by Rand's Project AIR FORCE.

interdiction, and electronic communication and jamming. TWIRL was developed to experiment further with the ROSS language and to provide a prototype simulation that could be used to explore issues in electronic combat.

Computerized simulation can be a notoriously expensive tool, consuming huge amounts of computer time on powerful machines. Worse, a simulation may require many runs to adequately sample and explore the simulated system's behavior. However, the speed of almost all simulations can be dramatically increased by exploiting their inherent concurrency. In 1981, Henry Sowizral and Dave Jefferson (then at USC, but a full-time Rand consultant that summer), began investigating the use of parallel processing in simulations. Their effort resulted in the design of a parallel processing computer architecture called Time Warp, which uses distributed simulation to significantly improve performance.[8] The Time Warp mechanism [47] accelerates a simulation's execution by removing the common restriction that simulated objects must always be synchronized in time.

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[8]Time Warp research has been supported by Rand's Project AIR FORCE, the System Development Foundation, and Rand corporate funds.

V. TECHNIQUES FOR EXPERT-SYSTEMS DEVELOPMENT

To be effective, expert systems must contain a substantial amount of domain expertise organized for efficient search. In the late 1970s, Rand began to address issues of acquiring and effectively using such expertise. Research was conducted in a number of areas, such as acquiring knowledge by example, iteratively refining and expanding that knowledge, devising efficient knowledge representations and structures for AI-based systems, and effectively using uncertain and inconsistent knowledge. During this same time period, work in opportunistic planning made extensive use of research in knowledge representation and organizational techniques to develop a cognitive model of planning [37,38].[9]

Rand's exemplary programming project focused on acquiring knowledge by example. The result of this work was the Exemplary Programming (EP) system [23,118], described in Chapter 8, which generates personalized computer programs from examples.[10] These programs can act as interfaces to complex computer systems or as intelligent assistants, freeing users from repeating detailed interactions with applications programs. Writing such programs often cannot be justified because of the large number of programs needed, their personalized nature, and their fast-changing specifications. However, the EP methodology provides quick, easy, and inexpensive methods for creating individualized software of this type.

The acquisition, expansion, and refinement of knowledge was the focus of a Rand project on machine-aided knowledge acquisition [41,69,40]. This project, described in Chapter 9, addressed the transfer of expertise from humans to machines, as well as the functions of planning, debugging, knowledge refinement, and autonomous machine learning. The relative advantages of humans and machines in the

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[9]This work was funded by the Office of Naval Research.

[10]EP was supported by the Defense Advanced Research Projects Agency and Rand corporate funds.

building of knowledge-based systems were also considered, along with issues of representing and structuring knowledge efficiently [62].[11]

A primary focus of Rand's research program has been the development of AI technology and its use in problem domains of practical importance. An important concern is the use of uncertain and inconsistent knowledge in expert systems. To study the effectiveness of inference methods under uncertainty, Ross Quinlan developed the INFERNO system [93].[12] This work, described in Chapter 10, documents the methods that today's expert systems use to grapple with inexact but valuable knowledge and suggests a new approach that avoids some of the problems.

Since 1979, Rand has conducted research in the area of distributed artificial intelligence [123,113,108,12,110].[13] Rand researchers have concentrated on developing and testing cooperative behaviors for air fleet control. Their work, described in Chapter 11, involves a network of cooperating expert systems and focuses on cooperative behaviors for plan generation, communication management, role negotiation, and data fusion.

The chapters in this book discuss much, but not all, of Rand's work in AI and expert systems over the last decade. This chapter and the technical chapters that follow give references and pointers to other relevant material. Nevertheless, the work presented here does highlight the major themes of Rand's continuing research on AI and expert systems.

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[11]These efforts were supported by a grant from the National Science Foundation.

[12]INFERNO was supported by Rand corporate funds.

[13]Rand's research on distributed problem solving has been supported by the Information Processing Techniques Office of the Defense Advanced Research Projects Agency.

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